algorithm must have explored some sequence which exposes a race, which by Claim 7.10 causes the algorithm to include a leaf with properties that contradict the initial assumption in the inductive step, thereby concluding the proof of the theorem.

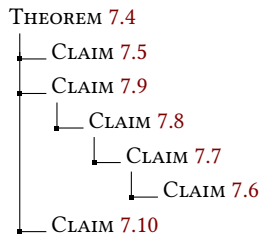

Fig. 8. Structure of correctness proof for Algorithm 2.

Theorem 7.4 (Correctness of Optimal-DPOR). *Whenever a call to Explore(E, Sleep, WuT) returns during Algorithm 2, then for all maximal execution sequences $E.w$, the algorithm has explored some execution sequence E' which is in $[E.w]_{\simeq}$.*

Since the initial call to the algorithm uses the arguments *Explore*($\langle\rangle, \emptyset, \langle\{\langle\rangle\}, \emptyset\rangle$), Theorem 7.4 implies that for all maximal execution sequences E the algorithm explores some execution sequence E' which is in $[E]_{\simeq}$. Since any execution sequence can be extended to a maximal one, the theorem also implies the analogue of Theorem 5.2, namely that for all execution sequences E, Algorithm 2 explores some execution sequence E' which is in $[E.v]_{\simeq}$ for some v.

Proof. By induction on the set of execution sequences E that are explored during the considered execution, using the ordering $\propto$ (i.e., the order in which the corresponding calls to *Explore* returned).

Base Case: This case corresponds to the first sequence E for which the call *Explore*($E, \cdot, \cdot$) returns. By the algorithm, E is already maximal, so the theorem trivially holds.

Inductive Step: Let us assume that there exist values E, *Sleep* and *WuT*, such that when the call to *Explore*(E, *Sleep*, *WuT*) returns, there is a maximal sequence $E.w$ such that the algorithm has not explored any execution sequence E' in $[E.w]_{\simeq}$. We will show that this leads to a contradiction.

For such w to exist, E cannot be maximal, so *final_sleep*(E) contains at least one process. For $p \in$ *final_sleep*(E), define:

- E'_p, such that $E'_p \leq E$, $E'_p.p \in \mathcal{E}$, and $E'_p.p$ is the last execution sequence of this form that precedes E (w.r.t. $\propto$). If $p \in$ *done*(E) then $E'_p = E$, but if $p \in$ *Sleep* then E'_p is a strict prefix of E.
- w'_p, as $E = E'_p.w'_p$. It follows that $E'_p \vdash p \diamond w'_p$.

Inductive Hypothesis: The theorem holds for all execution sequences E' with $E' \propto E$.

Claim 7.5. $WI_{[E]}(w) \cap$ *final_sleep*$(E) = \emptyset$.

Using the terminology of source sets, this claim states that w is a counterexample to the statement that *final_sleep*(E) is a source set for the set of sequences w with $E \vdash w$.

Proof. By contradiction. Assume that there is a $p \in WI_{[E]}(w)$ with $p \in$ *final_sleep*(E). Since w is maximal, $p \in WI_{[E]}(w)$ implies $p \in I_{[E]}(w)$, therefore from Definition 4.1(1) there is a w'' such that $E.w \simeq E.p.w'' \simeq E'_p.w'_p.p.w'' \simeq E'_p.p.w'_p.w''$. By the inductive hypothesis applied to $E'_p.p$, the